

Assignment 8

Class	SE CMPN	Vertical_Basket	PC_PCC		
Course Code	PCCE05T	Course	Data Base Management Systems		
Announcement Date	23 rd September	Submission Date	30 th September 2025		
Self-Declaration It took me _____ 1 _____ hour(s) and _____ 20 _____ minute(s) to solve the assignment, ensuring accuracy, completeness, and adherence to professional and ethical standards.					
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Assessment					
		Faculty Signature			

Q1. Given a relation schema $R(A, B, C, D, E)$ with functional dependencies:

A.>

1. $A \rightarrow B$
2. $B \rightarrow C$
3. $CD \rightarrow E$
- a. Find the attribute closure of A, CD, and AB.
- b. Determine all the candidate keys.

B.> Suppose you're given that the closure of attribute set $\{X\}$ is $\{A, B, C, D, E\}$. Propose at least two different sets of functional dependencies that might lead to this closure.

Q2. Given:

$R(P, Q, R, S)$ and $F = \{P \rightarrow Q, PQ \rightarrow R, R \rightarrow S\}$

- a. Compute P^+
- b. Construct a minimal cover for F

Q3. A relation $R(\text{StudentID}, \text{CourseID}, \text{Instructor}, \text{Grade}, \text{Department})$ has the following dependencies:

$\text{StudentID}, \text{CourseID} \rightarrow \text{Grade}$

$\text{CourseID} \rightarrow \text{Instructor}$

$\text{Instructor} \rightarrow \text{Department}$

1. a. Identify the anomalies in this schema.
- b. Normalize the schema into **3NF**.
- c. Can you further normalize it into **BCNF**? Justify your answer.

Q4. Given the following scenario:

A university stores data in a relation $R(\text{StudentID}, \text{Name}, \text{CourseID}, \text{CourseName},$

InstructorName).

- a. Assume appropriate dependencies and normalize the relation to **BCNF**.
- b. Suggest a real-world situation, where staying in **3NF** might be preferable to BCNF and explain why.

Q5. Create a relation schema with at least four attributes and a set of functional dependencies such that it:

- Is in 1NF
- Violates 2NF and 3NF

Then: Normalize it step-by-step to BCNF with explanation at each step.

Q6. Design a relation *R* with a set of FDs such that decomposition into BCNF is **lossless but not dependency-preserving**. Explain the trade-off between normalization and dependency preservation with reference to your example.

Q7. A hospital maintains a table:

Patient(PatientID, Name, DoctorID, DoctorName, Department, RoomNumber)

- A doctor works in only one department
- A patient is assigned one doctor, but multiple patients may be assigned the same doctor
- Patients can share room numbers

Tasks:

- a. Identify the functional dependencies
- b. Normalize the table to BCNF
- c. Discuss what happens to **data redundancy** and **query complexity** after normalization